

Srinath Jaina

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SUMMARY

- **Highly skilled Engineer with extensive experience (4+ years) in both front-end and back-end development.**
- Effectively managed **concurrent deliverables** for two projects, Multi Cloud Provisioning Service and Accelerite Cloud Platform projects.
- 4+ years of **professional** IT experience in Project analysis, Design, Development, Testing and Deployment of Web Based Applications using **Golang**. Actively involved in each phase Software Development Life cycle (**SDLC**)
- Proficient in utilizing **Go** frameworks such as **Gin, Echo, Fiber, Gorilla/mux, gRPC-go, Go kit**, and **Ginkgo, Gomega** testing frameworks for building efficient and scalable web applications and **microservices**.
- Engineered **Python** scripts for integration and functional testing of **Rovius Cloud Platform** powered by **Apache CloudStack**
- Specializing in **network security** and **VPN** technologies. Proven ability to design, develop, and maintain secure and high-performance **network** applications
- **Database**: Worked on both SQL and NoSQL databases like **MySQL, MongoDB, REDIS, DynamoDB, PostgreSQL**
- Expertise in implementing **proxy, forwarding, and tunneling protocols**. Strong understanding of **encryption** algorithms and security best practices. Passionate about building robust and resilient **network infrastructure**
- Experience with **Docker** Containers, leveraging Linux **AMI's** to create **Docker** Images/containers.
- Experience in Software Development Life Cycle (**SDLC**) with Architecture, Design, development, testing, implementation, maintenance & support for small, medium, and large-scalable systems.
- Expertise in advanced software development methodologies such as **Agile/Scrum** processes
- In-depth expertise in **cloud platforms (AWS, Azure, GCP, CloudStack)**, including **designing and implementing virtualization solutions** (leveraging both bare metal and hypervisors like **Xenserver, KVM, VMware**)
- Experience integrating applications with advanced application monitoring tools like **Prometheus, Datadog, ELK etc.**
- Proficient in configuring and deploying cloud platform using Amazon Web Services (AWS technologies like **EC2, S3, RDS, DynamoDB, Lambda, EMR, API Gateway, SNS, SQS and SES**)
- Proven experience in **Linux** system administration, including **RHEL/CentOS** and **Ubuntu** environments.
- Additionally, possess expertise in **Terraform**, enabling it to automate infrastructure provisioning and management across various cloud platforms like **AWS, GCP, Azure, OpenStack, Apache CloudStack**
- Architected and maintained **CloudStack** environments, provisioning **Virtual Machines, Networks, Kubernetes** clusters, and associated resources. Implemented and managed network security infrastructure, comprising **firewalls, routers, switches, and VPNs**, to safeguard systems and data.
- Deployed and managed **Kubernetes** clusters in both on-premises and cloud environments. Hands-on experience with **Kubernetes**

EDUCATION

Masters in Computer Science

University of Central Missouri • US • May 2024

SKILLS

Programming Languages: - Golang, Java, Python, Bash, C, C++

Operating Systems: - Linux, Unix, Windows

Golang Frameworks: - Gin, Echo, Beego, Ginkgo

Golang Libraries: - Gorilla/mux, GORM, gRPC-Go, Go networking libraries (net, net/http), OpenVPN, WireGuard, Cobra, viper, Gorilla Mux, go-mega

Protocols: TCP/IP, UDP, HTTP/HTTPS, TLS, DTLS, IPSec, IKEv2

Security: Encryption (AES, RSA), Authentication (OAuth2, JWT), Firewall, IDS/IPS

Cloud Environments:- AWS, GCP, AZURE, RoviCloud Platform.

Infrastructure as Code/Hashicorp Tools: Terraform, Ansible, Packer, Nomad, Vault

Databases: - MySQL, PostgreSQL, MongoDB

Monitoring tools: - Datadog, Prometheus, Elastic search

DevOps and Containerization: Docker, Kubernetes, CI/CD Pipelines and automation frameworks, Jenkins, AWS DevOps(Cloud Watch, S3 Bucket, DynamoDB, Lambda, SNS, SQS, DLQ, Cloud Formation, IAM Roles and Policies)

EXPERIENCE

Software Engineer, Persistent Systems Limited. • Aug 2018 – Aug 2022

Multi Cloud Service

- Architected and implemented a **Golang**-based **API** platform using the **Gin** framework and **Gorm** ORM to provision cloud resources across multiple public and private cloud environments. Utilized **HashiCorp's Terraform, Packer, Nomad, and Vault** to automate infrastructure deployment and management.
- Leveraged **gRPC** for efficient inter-service communication and implemented **Goroutines** to facilitate **concurrent job** execution.
- Implemented query functionality while mapping from end code to backend by a priority of native SQL, HQL, and **Gorm** Queries. Have been involved in implementing the DAO layer using **GORM** API.
- Worked on the **Go** Language collection types such as Slices, Structs, Maps, Interfaces, Goroutines and Channels.
- Using **GO**, developed a microservice for reading large volumes of data from **PostgreSQL** databases.
- Enhanced, fixed functional and performance related bugs in core products written in **GO**.
- Designed configurable **REST API's** on demand using **Go**. Worked on numerous go frameworks like **Gin, Revel, Beego**.
- Led the development of a high-performance **VPN** solution using **Golang**, implementing advanced features like **split tunneling, obfuscation, and multi-hop routing**.

- Designed and implemented a secure **proxy** server for filtering and **routing network** traffic, ensuring compliance with security policies and protecting against **malicious** attacks.
- Developed custom **network protocols** for optimized data transmission and secure communication between endpoints.
- Integrated strong **encryption** and **authentication** mechanisms into network applications to safeguard sensitive data.
- Conducted comprehensive **security** audits and **penetration testing** to identify and mitigate vulnerabilities.
- Optimized network performance by fine-tuning TCP/IP parameters and implementing efficient congestion control algorithms.
- Created **RPC**-based service for reading large volume of **data** from **Oracle** server using **Go**
- Used Testing tools like **Ginkgo** and **Go mega framework**.
- Managed **AWS, AZURE, GCP, and Cloudstack** Infrastructure using **Terraform**, Implemented a consistent workflow in **Golang** to provision and manage infrastructure throughout its lifecycle.
- Designed, implemented and maintained **terraform** scripts to automate infrastructure provisioning across various **public clouds**, including **AWS, AZURE, and GCP**.
- Developed a local **terraform** registry in **Golang** to enable the provisioning of resources to Accelerite private Cloud Platform (Rovius CP).
- Implemented **Swagger** for API documentation and added an **error**-handling mechanism in **Golang** to the Provisioning service.
- Leveraged **Postman** extensively for **API testing**, fostering **collaboration** by sharing test collections across UI, Billing, and DevOps teams
- Implemented various screens for the front end using **ReactJS**, Used **React** - Router to turn the application into single page Application.
- Deployed microservices in **Docker** containers, orchestrated containerized applications using **Kubernetes** which reduced the deployment times by 50%
- Integrated **Kafka** for real-time data streaming, ensuring seamless data flow within the pipeline.

Accelerite Cloud Platform

- Participated in developing new features based on customer requirements both in Backend (**JAVA**) and FrontEnd (**JavaScript, Angular**).
- Developing Web API using **Java8/J2EE, JSP, Servlets, JDBC, Spring, SOAP, REST, HTML and XML**.
- Utilized Spring Boot to modularize the code and implement Rest **API** using Spring **Rest** and **Swagger** API for documentation purposes
- Developed applications using **J2EE** technologies like **Hibernate, Servlets, and JSP**
- Built cloud Microservices and implemented back-end business logic using **Spring Boot**
- Support and maintenance of a private cloud **Accelerite Cloud Platform** which was based on **Apache Cloudstack** (Review **bugs** and problems as assigned, **diagnose** and fix them in an appropriate manner).
- **Deployed, Supported, and Managed** private **cloud** environment built on Open Source Cloud platforms like **Apache CloudStack** seamlessly integrating **cloud development** practices
- Developed **Shell** scripts for automating the **IaaS** management server installation and configuration, reducing manual efforts by **85%**.
- Enhanced **automation** framework on **Jenkins** and fixed **test scripts** on **GitLab** decreasing

90% test failures due to framework issues.

- Led regular **Build** Verification Tests and Regression tests, **troubleshooting** critical issues, which reduced post-release defects.
- Deployed **Docker** containers on **Linux VMs** and exhaustively tested **Kubernetes cluster** deployment in **CloudStack**.
- Designed and developed **scripts** for bulk operations on **CloudStack**. Executed performance tests discovering **96%** faster **API** response time and improved **CPU** and **memory** usage.
- Authenticated users and provided security measures to access **RESTFUL** services using **OAuth 2.0**
- Manual **network** configuration, including IP addressing, security, and firewalls.
- Managed and maintained a complex network infrastructure using **VMware**, ensuring high availability and performance
- Created and managed Virtual Machines using **Xenserver XenCenter**, **VMware vCenter** server appliance.
- Conducted a proof of concept (POC) to assess **MySQL** database high availability and reliability, Orchestrating the installation and configuration of **MySQL NDB cluster** for the Accelerite **Cloud Platform**.
- Developed and deployed a **trial user authentication** system for the Accelerite Cloud Platform, enabling users to access the product for a limited time, thus facilitating seamless trial experiences and fostering user engagement.
- Enhanced the capabilities of the Accelerite Cloud Platform by integrating support for **XenServer 8.2** hypervisor, ensuring compatibility with a broader range of **virtualization** environments and empowering users with increased flexibility in deployment options.
- Extensive experience in designing and implementing user interfaces (UI) for web applications using modern JavaScript frameworks such as **React.js** and **Vue.js**, Proficient in creating interactive and responsive **UI** components, optimizing performance, and ensuring cross-browser compatibility
- Automated the CI/CD pipeline using **Jenkins**, **Git**, and **Maven**, reducing build time by 40% and increasing productivity.
- Developed and maintained **Bash** scripts for process automation, Configured and managed cron jobs to schedule automated tasks and maintenance activities.
- Worked on **Maven** to build the applications. Used **Log4j** for logging errors, messages and performance logs.
- Debugged and fixed both front-end and back-end bugs.
- Dealt with **Jira** as ticket tracking and workflow tool. Used **Confluence** tool for documentation.
- Documented all the features that are developed in **Atlassian** Tool.
- Hands-on experience on **GIT/GITLAB/BITBUCKET** to maintain the code base, Build and maintain **SQL** scripts, indexes, complex queries for data analysis and extraction.

Persistent Systems Limited. • July 2018 – Dec 2018

Software Intern

- Primarily worked on **data** preprocessing and **data** cleaning, then selecting the right **ML** model and improved its accuracy to 90 %.
- Built **Restful API's** using **Python Flask** to enhance the communication between UI in which the users upload the pictures of their damaged vehicle, and these images are sent to the machine learning algorithm which essentially processes these images received and validates the proximity of the damage and evaluates the amount of insurance that can be claimed by the user. Contributed in developing the machine learning algorithm as well
- Hands-on experience on **GITLAB** to maintain the code base.
- Worked on front-end assignment projects using UI framework **AngularJS**.